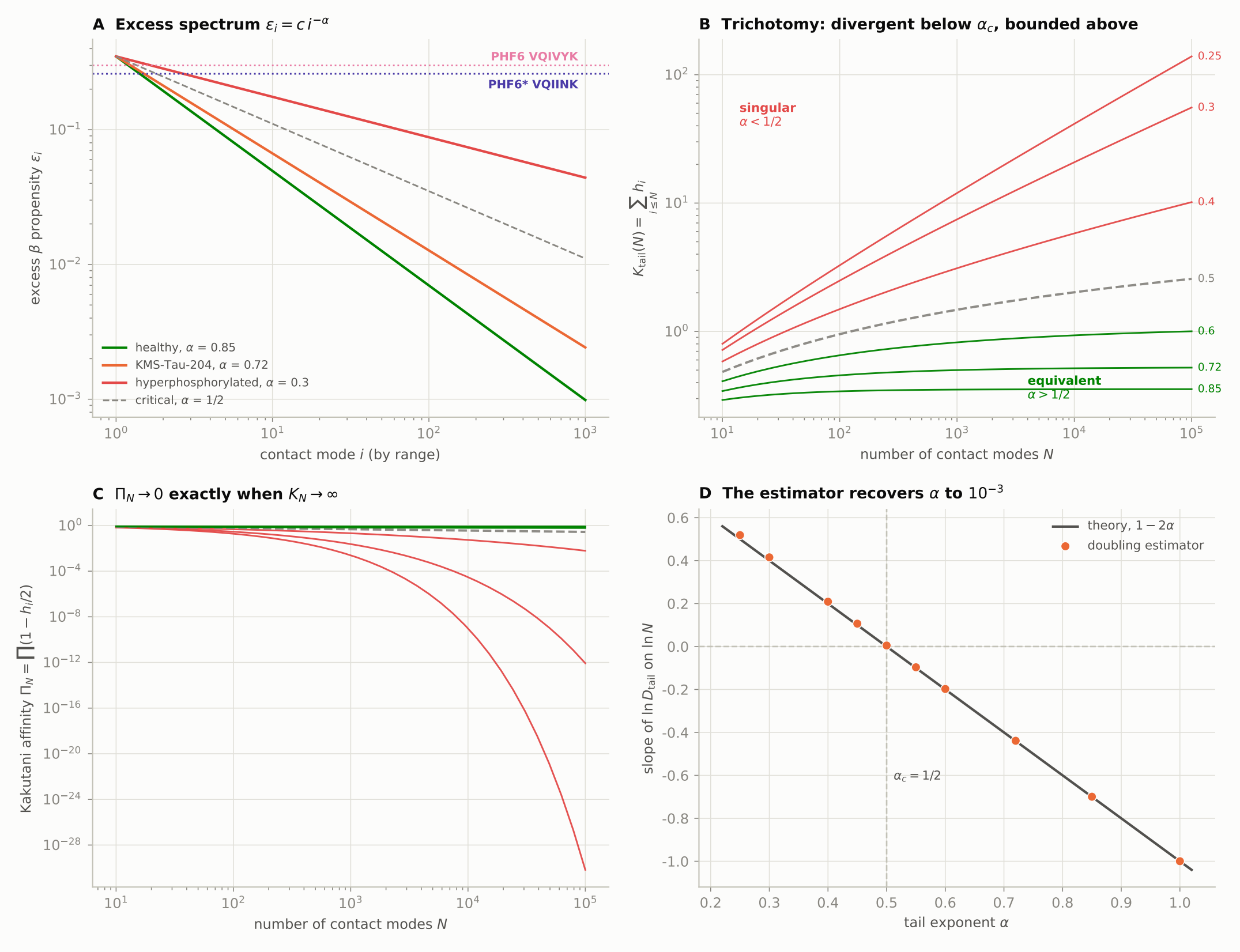
A Excess spectrum $\varepsilon_i = c\, i^{-\alpha}$
PHF6 VQIVYK
PHF6* VQIINK
healthy, α = 0.85
KMS-Tau-204, α = 0.72
hyperphosphorylated, α = 0.3
critical, α = 1/2
excess β propensity ε_i
contact mode i (by range)
B Trichotomy: divergent below α_c, bounded above
singular
$\alpha < 1/2$
equivalent
$\alpha > 1/2$
0.25
0.3
0.4
0.5
0.6
0.72
0.85
$K_{\mathrm{tail}}(N) = \sum_{i \le N} h_i$
number of contact modes N
C $\Pi_N \to 0$ exactly when $K_N \to \infty$
Kakutani affinity $\Pi_N = \prod(1 - h_i/2)$
number of contact modes N
D The estimator recovers α to 10^{-3}
theory, $1 - 2\alpha$
doubling estimator
$\alpha_c = 1/2$
slope of $\ln D_{\mathrm{tail}}$ on $\ln N$
tail exponent α